2026 RICHFIELD HACKATHON GUIDELINES

Team Size	7 to 10 students per group
Date	TBC
Presentation	30 to 45 minutes, including a PowerPoint presentation and a live demo

Hackathon Brief

1. The Scenario

As graduate employability becomes increasingly competitive, professionals are expected to build strong digital portfolios, establish meaningful professional networks, and showcase their skills long before they graduate. At the same time, organisations are actively seeking graduates who demonstrate not only technical competence but also a visible professional presence and the ability to engage with industry.

To address this opportunity, Richfield is launching this Hackathon to challenge students to design and develop a real-world digital solution while strengthening their own software development skills. The hackathon provides students with the opportunity to apply the knowledge gained throughout their qualification by designing, developing, testing, and presenting a complete software application that addresses an authentic institutional need.

Your challenge is to design and build a professional networking mobile application for the Richfield/AAA community. The platform should connect current students, alumni, recruiters, and industry partners within a secure and trusted ecosystem. It should enable students to establish

professional profiles before graduation, showcase their skills and achievements, expand their professional networks across campuses, and continue growing their careers as alumni.

The platform should also provide recruiters and industry partners with an efficient way to discover Richfield talent, advertise opportunities, engage with graduates, and participate in career-related events. By creating a dedicated professional community, the application has the potential to strengthen collaboration between students, alumni, academia, and industry while enhancing graduate visibility and long-term career development.

This hackathon is more than a software development competition, it is an opportunity to solve a meaningful real-world challenge, demonstrate full-stack application development skills, strengthen teamwork and innovation, and contribute towards a digital platform that could support the Richfield community for years to come.

2. The Project Brief

Your team must build a mobile application. All requirements listed in this section are mandatory. A prototype that omits any of these requirements will be assessed as incomplete, regardless of the quality of what is present.

2.1 User Types and Authentication

The platform must support four distinct user types, each with a different role, access level, and authentication method. Role-based access control must be enforced on the backend, hiding screens on the front end alone is insufficient.

- Student: currently enrolled at Richfield. Must authenticate using their Richfield institutional email address (@my.richfield.ac.za, @richfield.ac.za, @my.aaa.ac.za or @aaa.ac.za). Registration must be restricted to this domain.
- Alumni: graduates of Richfield who may no longer have active access to their student email accounts. Teams must research, design, and implement a credible alternative authentication and identity verification flow for this user group. This is a deliberate design challenge; there is no prescribed solution.

- Business User: a verified employer or recruiter. Registration requires a separate approval or verification step before access is granted. Business users may not access student-only content or personal information beyond what a student has explicitly made visible.
- Administrator: a Richfield staff member with elevated access to manage the platform. Administrators do not use the same registration flow as other user types; admin accounts must be provisioned securely and separately. Administrator functionality is described in section 2.2.

NB:

Hard-coded credentials, mock logins, or placeholder authentication will not be accepted for any user type.

Recommended tools include Firebase Authentication, Supabase Auth, Auth0, or a custom JWT-based implementation.

Your team must justify the authentication approach chosen for each user type in the presentation. The alumni authentication challenge must be addressed with a researched, reasoned solution, not left as a gap. Administrator accounts must never be self-registerable; the provisioning and protection of admin credentials must be demonstrated and explained.

2.2 Administrator Panel

The administrator is a Richfield staff member responsible for the platform's health, integrity, and operation. The admin panel does not need to be a mobile screen, it may be a separate web-based or desktop management interface, but it must be functional and demonstrable.

- User management: the administrator can view, approve, suspend, or remove any user account. This includes approving business user registrations and verifying alumni identity where required
- Content moderation: the administrator can remove posts, videos, or comments that violate platform guidelines, and can flag accounts for review

- Event management: the administrator is the only user type who can create, edit, and publish official Richfield institutional events (career fairs, workshops, networking events). These events appear in the event listings visible to all users.
- Opportunity oversight: the administrator can review, approve, or remove opportunity listings posted by business users before they become visible to students
- Platform analytics: the administrator has access to a high-level platform dashboard showing total active users by type, registration trends, content volume, engagement rates, and flagged content requiring review
- Announcement broadcasting: the administrator can push platform-wide announcements or notices to all users or to specific user groups

2.3 User Profiles

All user types must have a profile. Profile functionality must be comprehensive and professional, serving as a digital portfolio rather than simply displaying a user's name and photograph.

- **Student and alumni profiles** must include:
    - Full name and profile photograph
    - Professional headlines and short professional summary
    - Programme of study (students) or current field of work (alumni)
    - Campus attended
    - Year of enrolment and graduation (where applicable)
    - Technical and professional skills
    - Work experience, internships, and industry placements
    - Entrepreneurial experience, including businesses founded, start-ups, freelance work, innovations, or business ventures
    - GitHub profile link

    - A portfolio of GitHub projects with links to each repository
    - A portfolio of developed websites or applications with live URLs (where applicable)
    - List of digital badges earned
    - Professional certifications
    - Link to the user's Credly digital badge profile
    - Academic achievements, scholarships, awards, and recognitions
    - Leadership roles (e.g., Student Representative Council (SRC), class representative, mentor, student ambassador)
    - Participation in student clubs, societies, entrepreneurship hubs, hackathons, competitions, innovation challenges, volunteer initiatives, and community engagement activities
    - Career interests and aspirations
    - Professional networking links (e.g., LinkedIn)
- **Students and alumni** may upload a curriculum vitae (CV), supporting documents, or provide links to external portfolio platforms such as GitHub, LinkedIn, personal websites, Behance, or other professional portfolios.
- **Business users** must maintain a company profile that includes the organisation name, industry, company description, location, company website, contact details, and the types of graduates, skills, or talent they are seeking.
- **Profile visibility controls** must allow users to manage which sections of their profile are visible to business users, fellow students, alumni, lecturers, and the public.
- **Endorsements and Recommendations:** Users must be able to endorse specific skills displayed on another user's profile. Alumni, lecturers, mentors, supervisors, and employers may also provide short written recommendations or testimonials that appear on the recipient's profile.

2.4 Connections and Social Features

Your app must support a professional connection model and an active content feed.

- Users can send, accept, decline, and manage connection requests
- A personalised activity feed displays posts, videos, and updates from a user's connections, ranked by relevance to that user's role and engagement history, the feed must behave differently for students, alumni, and business users
- Users can create text posts and share professional updates
- Students and alumni can upload short-form videos, day-in-the-life content, career journey stories, or campus experience footage, intended to inform and inspire current students in their career decisions
- Direct messaging, connected users can send private messages to one another
- Users can comment on and react to posts and videos

2.5 Opportunity Listings and Career Features

The platform must serve as a functional bridge between students and the job market.

- Business users can post internships, learnerships, part-time roles, and graduate vacancies, subject to administrator approval before listings go live
- Students can browse, filter, and apply for or express interest in posted opportunities directly through the app
- A career pathway explorer allows students to view the career trajectories of alumni who studied the same programme, showing where graduates have ended up and what roles they have held
- Institutional events created by the administrator appear in an event listings feed; students receive notifications for events relevant to their programme or interests

2.6 Onboarding, Chatbot, and Profile Assistant

Every new user must be guided through a structured onboarding experience upon first login. This is not optional it is a core feature of the platform.

- An in-app AI-powered profile assistant must guide new users step-by-step through completing their profile, explaining the purpose of each section and what makes it compelling to employers or connections
- The assistant must provide context-aware feedback for example, flagging that a profile with no skills listed is unlikely to appear in business user searches, or that a profile photo significantly increases engagement
- The assistant must also function as an ongoing resource: users can return to it at any time to ask questions about how to improve their profile, what types of posts attract employer attention, or how to approach making connections
- An interactive app tutorial must walk first-time users through the key screens and features of the app before they begin using it
- The chatbot and tutorial must be built using a recognised AI or conversational framework a hardcoded script of fixed responses is not sufficient

2.7 Data Analytics Dashboards

Data analytics must be implemented as three distinct dashboards one per applicable user type. A single generic statistics page shared across user types will not satisfy this requirement.

- Student analytics dashboard, must show profile view count and trends over time, connection growth, post and video engagement metrics, how their profile completeness compares to peers in the same programme, and which of their skills are most searched for by business users
- Business user analytics dashboard, must show total applicant pipeline per posted opportunity, engagement rate on job listings, candidate skill distribution across Richfield programmes, demographic breakdown of applicants by programme and year of study, and overall reach and visibility statistics for the company profile

- Administrator analytics dashboard must show: platform-level metrics including total registered users by type, monthly active users, new registrations over time, content volume (posts, videos, opportunities), flagged content count, and business user approval pipeline status

All three dashboards must present data visually using charts, graphs, or visual indicators. Raw numbers displayed without visualisation will not satisfy this requirement.

2.8 Additional Technical Features

The following technical features are mandatory across the entire application:

- Real-time notifications connection requests, new messages, opportunity matches, administrator announcements, and activity on a user's posts must trigger live push notifications or in-app alerts, implemented using WebSockets or a push notification service. Polling-based solutions are not acceptable.
- Smart job matching when a new opportunity is approved and published, the system must automatically surface it to students whose profile skills and programme align with the listing. This must be driven by matching logic, not a generic broadcast to all users.
- NLP-assisted profile building when a student uploads a CV or enters free-text into their profile, the system must use natural language processing to extract and suggest skills, qualifications, and experience entries automatically
- Video processing uploaded videos must be compressed or transcoded before storage. Thumbnail generation is required. Raw, unprocessed video uploads dumped directly to storage are not acceptable.

3. Technology Expectations

The following stack decisions are mandatory by category teams making the specific choice within each category based on their own research. Every choice must be documented and justified in the presentation.

3.1 Mobile Framework

The application must be built as a mobile application using one of the following frameworks. Teams may not submit a web application or desktop application as a substitute.

- Flutter with Dart: Google's cross-platform toolkit producing native iOS and Android apps from a single codebase
- React Native with JavaScript or TypeScript: Meta's framework for building native mobile apps using React
- C# with .NET MAUI: Microsoft's cross-platform framework for building native mobile and desktop apps
- Kotlin (Android native) or Swift (iOS native): accepted only where the team explicitly targets a single platform and provides a strong written justification

The choice of framework must be justified in the presentation. Teams must explain why their chosen framework was the most appropriate for this project, citing the platform's requirements and their team's technical context.

3.2 Database

The application must use a real, industry-standard database management system. Teams must research both SQL and NoSQL approaches and make a deliberate, justified choice. The decision must be documented in the presentation with reference to the platform's specific data requirements.

- SQL options: PostgreSQL, MySQL, Microsoft SQL Server
- NoSQL options: MongoDB, Firebase Firestore, Supabase

Teams must consider the nature of their data (relational vs. document-based), scalability, real-time query requirements, and compatibility with their chosen mobile framework. Selecting a database without demonstrating research will not attract marks for this decision.

4. Deliverables

Each team must submit and present the following by the agreed deadline. All deliverables must be ready before the presentation begins.

WORKING MOBILE PROTOTYPE

A functional, runnable mobile application demonstrating all mandatory features across all four user types. The prototype must operate on a physical device or emulator during the live demo. The administrator panel may be demonstrated separately if it is web-based. Wireframes, web applications submitted as the main app, or desktop applications will not be accepted as substitutes.

SOURCE CODE REPOSITORY

A Git repository (GitHub, GitLab, or Bitbucket) with a populated README file covering project overview, setup instructions, and technology choices. Commit history must reflect meaningful, distributed contributions from all team members throughout the project period. A repository with a single large commit or contributions from only one or two members will be treated as evidence of non-compliance.

POWERPOINT PRESENTATION

A professional slide deck of no fewer than 15 slides, covering all of the following:

- Problem statement and scenario response
- User type breakdown all four roles and their access levels
- Authentication design including alumni solution and admin provisioning
- System architecture diagram
- Database and mobile framework choices with justification

- AI profile assistant and chatbot design and implementation
- Data analytics dashboards separate slides for each user type
- Data privacy and POPIA compliance

LIVE PRESENTATION AND DEMO

The live presentation is 30 to 45 minutes, inclusive of a live working demo. All team members must present a portion of the material. The demo must show the platform operating for all four user types, including the administrator panel. Teams should plan their demo flow in advance and ensure it runs without critical failures; a demo that crashes or cannot be navigated confidently will affect marks under Presentation Quality. Narrate what you are demonstrating the panel should not have to guess what they are seeing or why it matters.